7. $\xi = 0$ (i)

$$\hat{H} = -\frac{J}{2}\sum_{pq} P_p^+ P_q^-$$

$$H = -\frac{1}{2}J\begin{bmatrix} 2 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 & 0 & 1 \\ 1 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 1 & 2 \end{bmatrix}$$

we want $H|\psi\rangle = E|\psi\rangle$ where E is smallest, this gives the ground state. $\langle\psi|\psi\rangle$ must also be 1. For this $|\psi\rangle$ is the superposition between all states.

$$|\psi\rangle = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$H|\psi\rangle = -3g|\psi\rangle$$

$$E_0 = -3g$$

We can also verify that this is the ground state numerically:

(ii) $|n_1 n_2 n_3 n_4\rangle$ $\quad X = \frac{1}{\sqrt{6}}$

$$C^{[1|234]} = \begin{array}{c} \\ |0\rangle \\ |1\rangle \end{array} \begin{array}{c} \begin{array}{cccccccc} |000\rangle & |100\rangle & |010\rangle & |110\rangle & |001\rangle & |101\rangle & |011\rangle & |111\rangle \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & X & 0 & X & X & 0 \\ 0 & X & X & 0 & X & 0 & 0 & 0 \end{bmatrix} \end{array}$$

$$C^{(1\,2\,1\,3\,4)} = \begin{array}{c} \\ |00\rangle \\ |10\rangle \\ |01\rangle \\ |11\rangle \end{array} \begin{array}{c} \begin{array}{cccc} |00\rangle & |10\rangle & |01\rangle & |11\rangle \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & X \\ 0 & X & X & 0 \\ 0 & X & X & 0 \\ X & 0 & 0 & 0 \end{pmatrix} \end{array}$$

$$C^{(1\,2\,3\,1\,4)} = \begin{array}{c} \\ |000\rangle \\ |100\rangle \\ |010\rangle \\ |110\rangle \\ |001\rangle \\ |101\rangle \\ |011\rangle \\ |111\rangle \end{array} \begin{array}{c} \begin{array}{cc} |0\rangle & |1\rangle \end{array} \\ \begin{pmatrix} 0 & 0 \\ 0 & X \\ 0 & X \\ X & 0 \\ 0 & X \\ X & 0 \\ X & 0 \\ 0 & 0 \end{pmatrix} \end{array}$$

$$C^{(1|234)} = C_1 = U_1 \Sigma_1 V_1^+$$

$$C_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Sigma_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$V_1^+ = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

$$\Sigma_1 V_1^+ = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} = C_1$$

Reshape into 4x4:

$$C_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & \end{bmatrix}$$

$$C_2 = U_2 \Sigma_2 V_2^{\dagger}$$

$$\Sigma_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$U_2 = \begin{bmatrix} 0 & 0 & -1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$V_2^{\dagger} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

$$\Sigma_2 V_2^{\dagger} = \begin{bmatrix} 0 & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{\sqrt{6}} \end{bmatrix}$$

Reshape into 6 x 2:

$$C_3 = \begin{bmatrix} 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\frac{1}{\sqrt{6}} \end{bmatrix}$$

$$C_3 = U_3 \Sigma_3 V_3^{+}$$

$$\Sigma_3 = \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{bmatrix}$$

$$V_3^{+} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U_3 = \begin{bmatrix} 0 & -\sqrt{\frac{2}{3}} \\ -\sqrt{\frac{2}{3}} & 0 \\ -\frac{1}{\sqrt{3}} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} \end{bmatrix}$$

$$U_4 = \Sigma_3 V_3^{\dagger} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^{[1]0} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$A^{[1]1} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$A^{[2]0} = \begin{bmatrix} 0 & 0 & -1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \end{bmatrix}$$

$$A^{[2]1} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$A^{[3]0} = \begin{bmatrix} 0 & -\sqrt{\frac{3}{2}} \\ -\frac{1}{\sqrt{3}} & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^{[3]1} = \begin{bmatrix} -\sqrt{\frac{3}{2}} & 0 \\ \end{bmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & -\frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A^{[4]1} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

$$A^{[4]2} = \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$|\psi\rangle = \sum_{\substack{s_1 s_2 \\ s_3 s_4}} A^{[0]s_1} A^{[1]s_2} A^{[2]s_3} A^{[3]s_4} |s_1 s_2 s_3 s_4\rangle$$

(iii) $\chi = 3$ for the middle bond.

This means that there are 3 possible states for the 2 sites to the left of the bond. since we

only have 2 pairs of fermions we have:

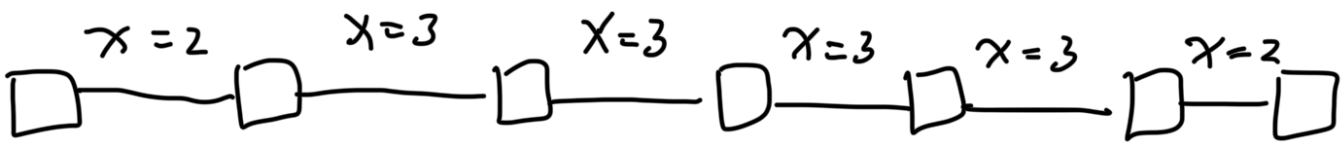
State 1: both sites occupied

State 2: no sites occupied

state 3: 1 site occupied

These are 3 possible bond states.

But one important thing to notice is that if we increase L to 5, both midde bonds will still have $\chi = 3$ because we are restricted to 2 pairs. this means that:



for any number of L.

So shapes of A matrices are (χ_{i-1}, χ_i):

$A^{[1]s_1} = (1 \times 2)$

$A^{[2]s_2} = (2 \times 3)$

$A^{[3]s_3}, \ldots A^{[L-2]s_{L-2}} = (3 \times 3)$

$A^{[L-1]s_{L-1}} = (3 \times 2)$

$A^{[L]s_L} = (2 \times 1)$

Constructing middle $A^{[i]s_i}$:

For this lets use bond indices $\alpha_i \in (0, 1, 2)$ which tells us how many piles have been

how many sites have already occupied.

$$A^{C_i|S_i}_{\alpha_{i-1},\alpha_i} = (3 \times 3) \text{ matrix with entries } \alpha_{i-1}, \alpha.$$

$A^{C_i|0}_{\alpha_{i-1},\alpha_i}$ must have entries $\alpha_{i-1} = \alpha_i$ to 1 and $\alpha_{i-1} \neq \alpha_i$ to 0 since $S_i = 0$.

$$A^{C_i|0} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$A^{C_i|1}_{\alpha_{i-1},\alpha_i}$ must have entries $\alpha_{i-1} = \alpha_i - 1$ to 1

and all others to 0.

$$A^{[i]1} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Close to the bond edges, all three bond states aren't Possible

8.

now $\xi = 1$ again and u